

WanderLens: A Camera-Roll Recommendation System

Predicting your next adventure

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The Paradox of Choice

The Search Struggle

Whether you're an experienced voyager or a new traveler seeking fresh adventures, finding relevant activities without endless searching is a common, time-consuming challenge.

Your Camera Roll, Our Key

Imagine a system that understands your unique preferences by analyzing the visual story of your camera roll, **transforming pixels into passions.**

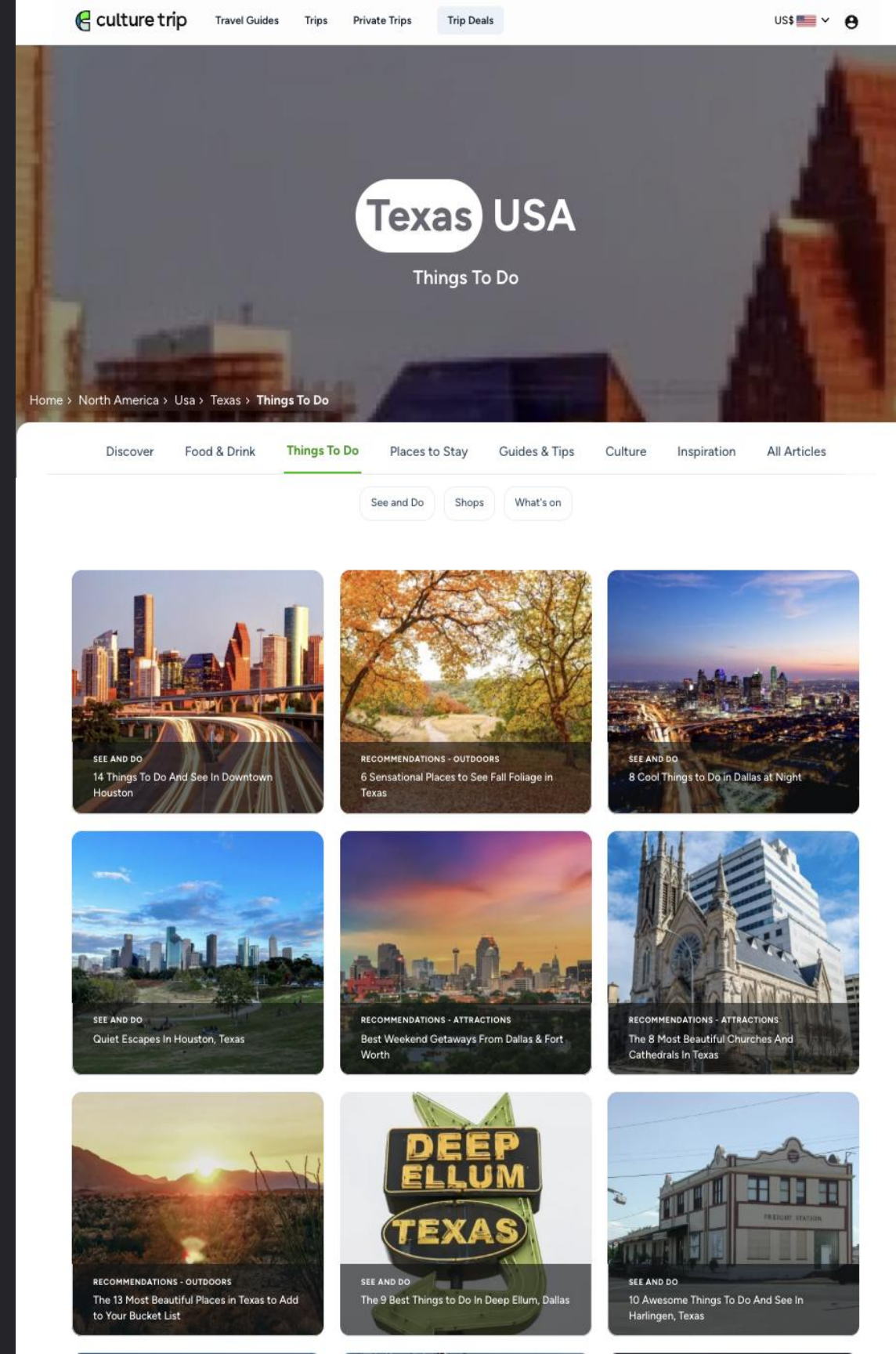
Instant, Personalized Fun

Get top 3 tailored recommendations in an instant, saving you time and potentially surprising you with new favorite spots based on your visual history.

Strategic Focus:

Data Sourcing Hurdles

Initial attempts to scrape diverse location data from major travel platforms (TripAdvisor, Viator, Booking, Kayak) presented significant technical challenges due to varied site structures.



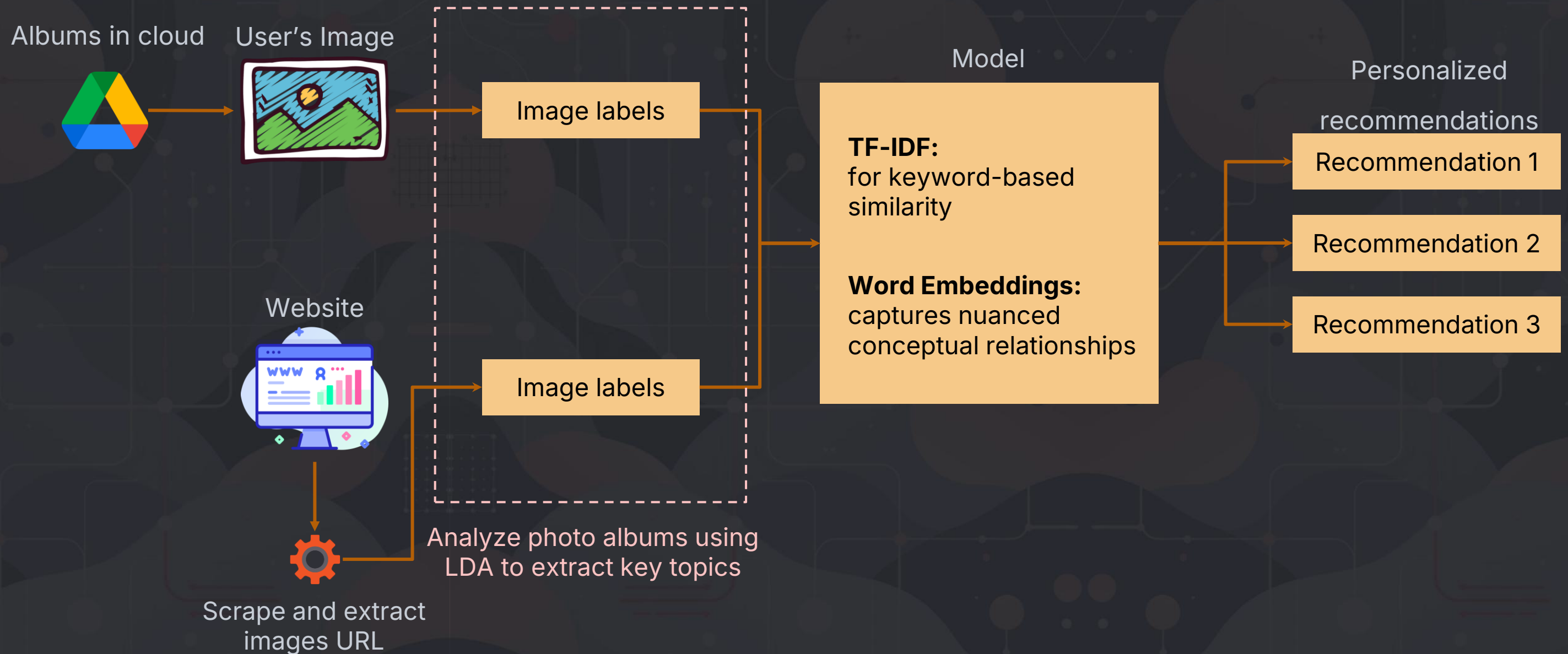
Why Texas?

Culture Trip Breakthrough

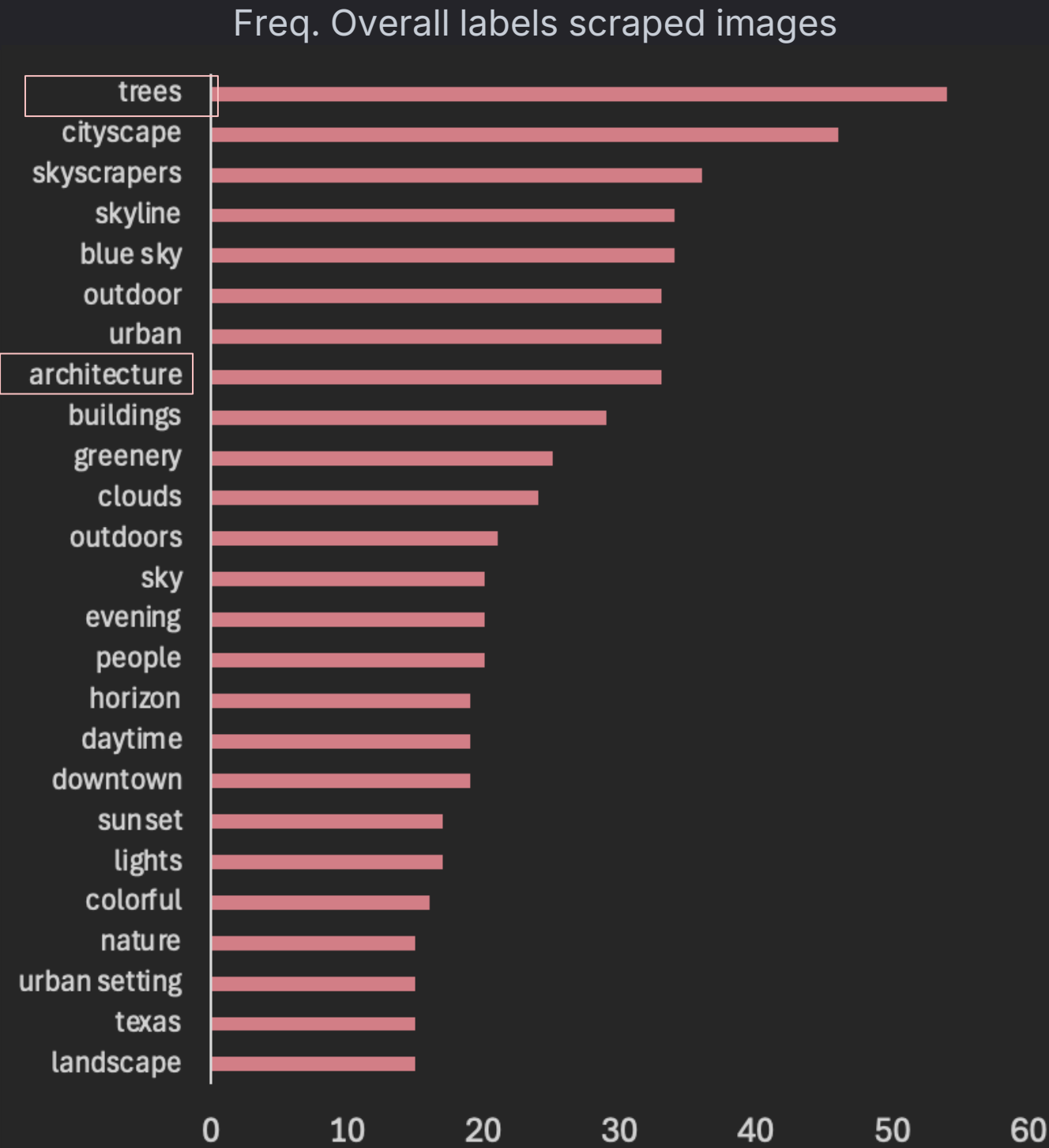
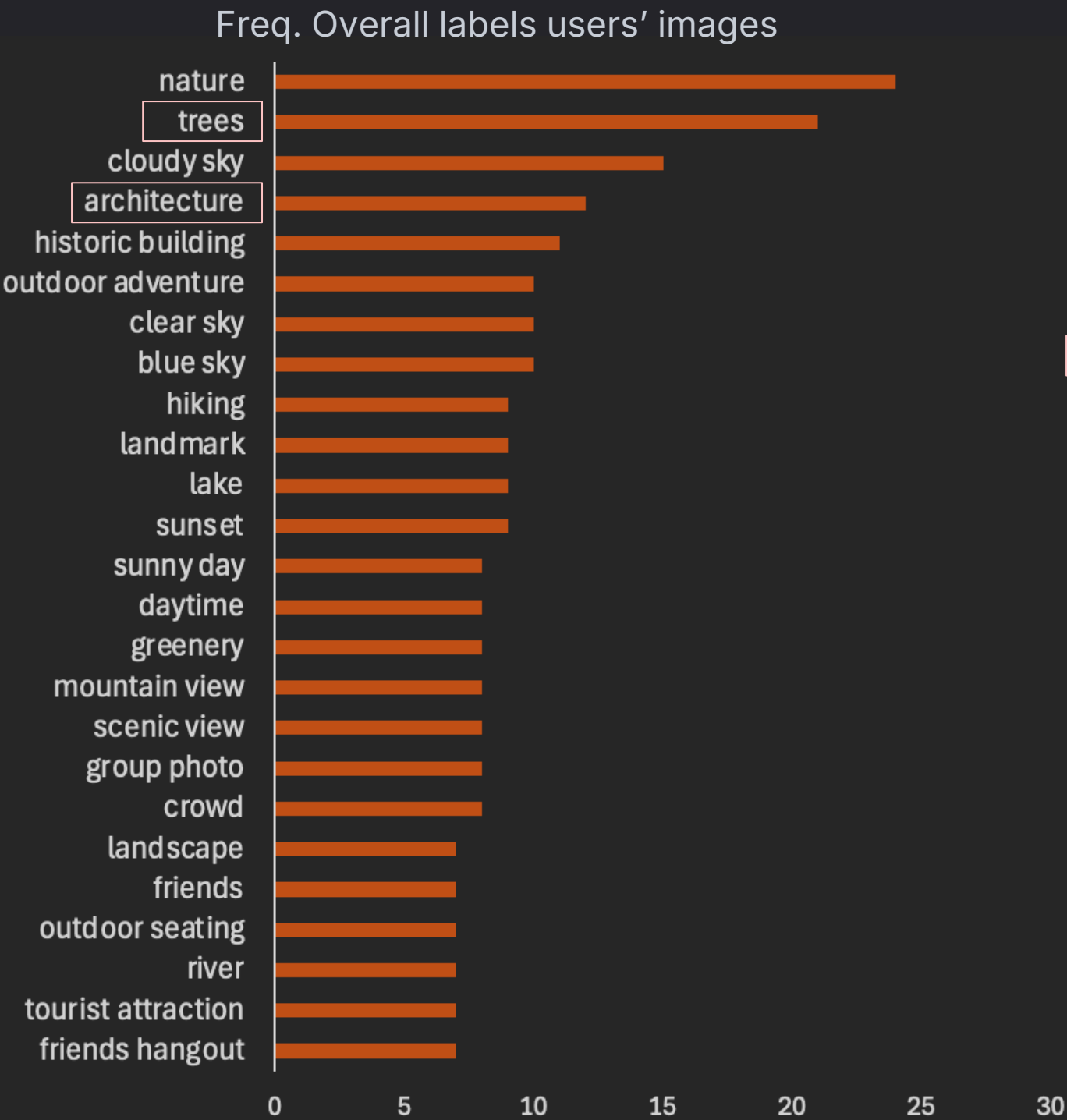
We successfully identified and utilized Culture Trip as a reliable data source, providing structured, high-quality content suitable for our location catalog.

This enabled us to build a rich dataset of 196 unique "Things To Do" in Texas, making it our ideal pilot region for validating the recommendation system.

Design & Implementation Details



Cosine Similarity



From your gallery to the next adventure



Spend A Night Under The Stars With Cinderella



<https://theculturetrip.com/north-america/usa/texas/articles/spend-a-night-under-the-stars-with-cinderella>



night, outdoor seating, night scene, seating, building

TF-IDF

From your gallery to the next adventure



Top Haunted Places in Texas for Would-Be Ghostbusters



<https://theculturetrip.com/north-america/usa/texas/articles/top-5-haunted-places-in-texas>

travel, sunset

TF-IDF

From your gallery to the next adventure



nature, trees, greenery,
forest, outdoors



Houston Landmarks For Science Lovers



<https://theculturetrip.com/north-america/usa/texas/articles/houston-landmarks-for-science-lovers>

WORD EMBEDDINGS

From your gallery to the next adventure

Enjoy Engineered Beauty At AIA Sandcastle Competition



beach, architecture, leisure



<https://theculturetrip.com/north-america/usa/texas/articles/enjoy-engineered-beauty-at-aia-sandcastle-competition>

WORD EMBEDDINGS

AHA MOMENT...

Summary

Takeaway #1

Our project successfully bridges the gap between unstructured visual data and actionable user insights, making AI accessible to everyone.

Takeaway #2

By leveraging state-of-the-art AI and recommendation techniques, we create a unique, engaging way to discover "things to do" that matters to you.

Takeaway #3

Unlock the hidden potential of your photos to enrich everyday experiences and create unforgettable memories.



Future Work: From Local Expert to Global Guide

Scaling the Model:

The Vision:

The successful Texas model serves as the blueprint. The next step is to expand its expertise.

The Path:

Ingest curated datasets for other regions (e.g., "Pacific Northwest," "New England") to create specialized guides for new locations, season, events, specific things, budget, travel transportation method.

The "Cold Start" Problem:

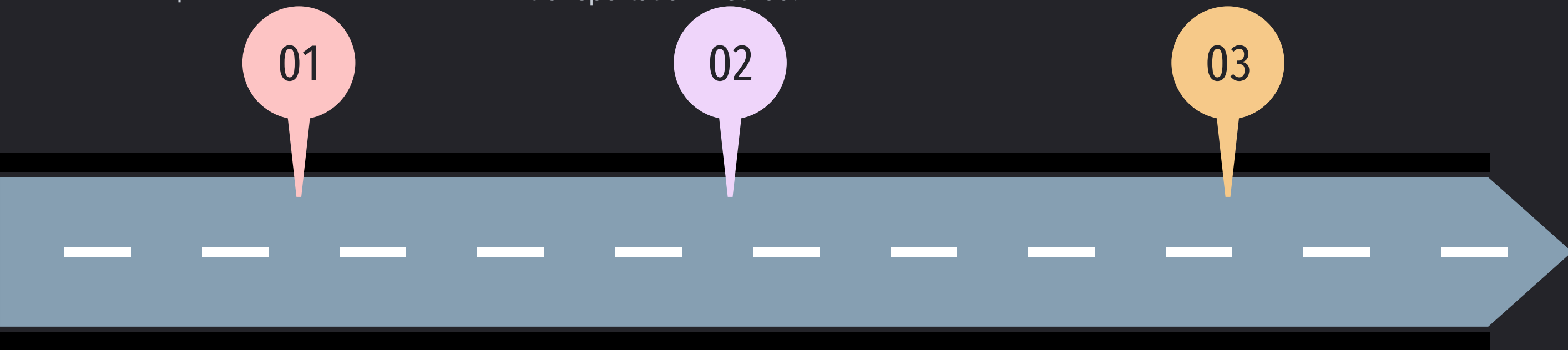
The Challenge: How do you recommend to a new user with no photos?

The Solution: Implement a simple onboarding quiz ("City nightlife or quiet nature trails?") to establish a baseline interest profile.

01

02

03



Thank you!

